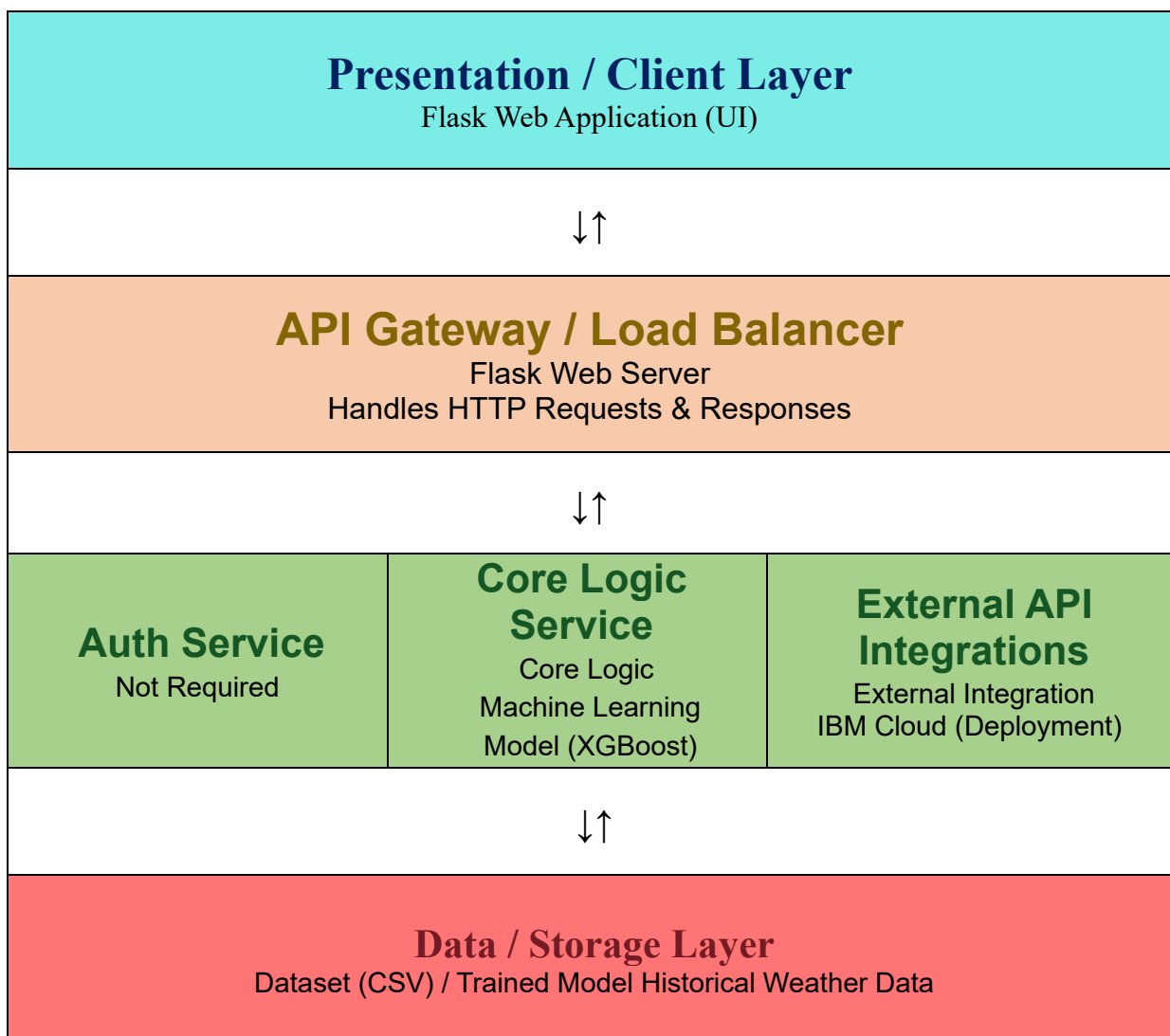


Solution Architecture

Date	29 June 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where users enter weather details and receive flood predictions.	Flask, HTML, CSS
API Gateway	Receives user input and sends it to the Machine Learning model for prediction.	Flask
Core Logic Service	Receives user input and sends it to the Machine Learning model for prediction.	Python, Scikit-learn, XGBoost, Pandas
Database / Storage	Stores the historical weather dataset and the trained Machine Learning model.	CSV Dataset, Pickle (.pkl)
Deployment Layer	Hosts the web application and enables online access with automatic deployment from GitHub.	Render, Gunicorn